

Internet of Things – Project 2

Automated Irrigation Controller (Actuator Logic)

Objective

To build an automated irrigation controller using an ESP32, a soil moisture sensor and a relay to automatically control a water pump based on soil moisture levels.

Implementation

- Read soil moisture through ESP32 ADC.
- Convert ADC value to moisture percentage.
- Use hysteresis (30% ON, 45% OFF).
- Control relay/LED.
- Display readings on Serial Monitor.

Source Code

```
const int soilPin = 34;
const int relayPin = 26;

const int ON_THRESHOLD = 30;
const int OFF_THRESHOLD = 45;

bool pumpState = false;

unsigned long previousMillis = 0;
const unsigned long interval = 2000;

void setup() {
  Serial.begin(115200);

  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, LOW);

  Serial.println("=====");
  Serial.println(" Automated Irrigation Controller");
  Serial.println(" DecodeLabs IoT Project 2");
  Serial.println("=====");
}
```

```

void loop() {
  unsigned long currentMillis = millis();

  if (currentMillis - previousMillis >= interval) {

    previousMillis = currentMillis;

    int moisture = map(sensorValue, 4095, 0, 0, 100);

    moisture = constrain(moisture, 0, 100);

    if (moisture < ON_THRESHOLD) {
      pumpState = true;
    }
    else if (moisture > OFF_THRESHOLD) {
      pumpState = false;
    }
    digitalWrite(relayPin, pumpState)
    Serial.println("-----");

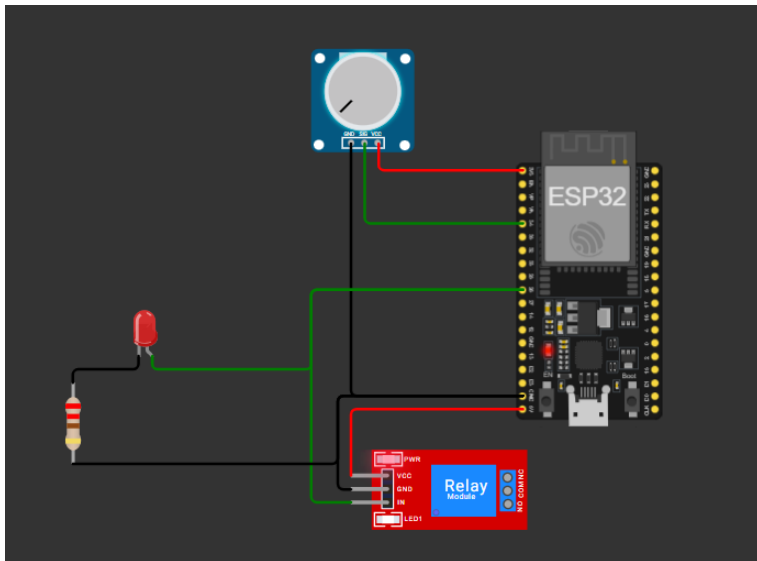
    Serial.print("ADC Value      : ");
    Serial.println(sensorValue);

    Serial.print("Moisture Percentage: ");
    Serial.print(moisture);
    Serial.println("%");

    if (pumpState) {
      Serial.println("Pump Status      : ON");
    } else {
      Serial.println("Pump Status      : OFF");
    }
    Serial.println("-----");
    Serial.println();
  }
}

```

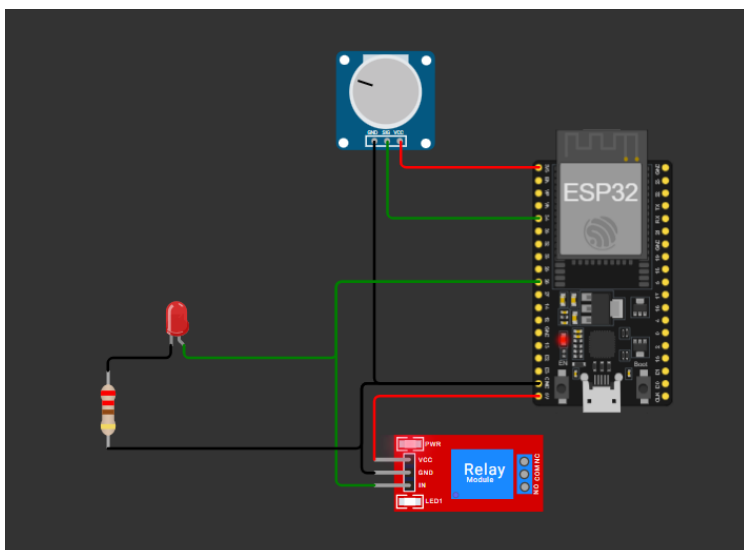
Results



```
=====
Automated Irrigation Controller
DecodeLabs IoT Project 2
=====

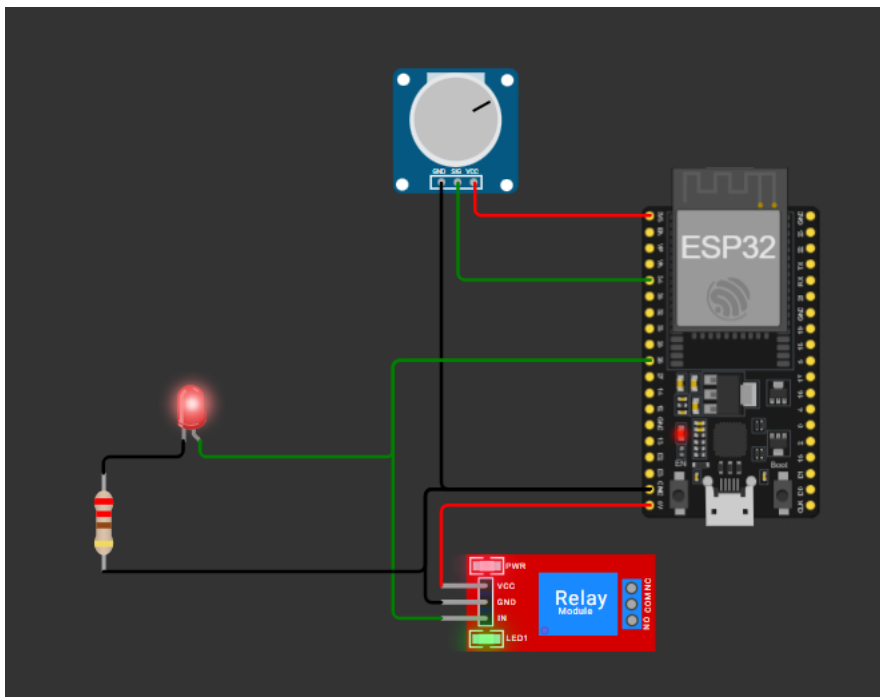
-----
ADC Value           : 0
Moisture Percentage: 100%
Pump Status         : OFF
-----
```

High moisture and low ADC value, Pump is in OFF State



```
-----  
ADC Value           : 957  
Moisture Percentage: 76%  
Pump Status         : OFF  
-----
```

Moderate Moisture and ADC less than 1000, Pump is in OFF State



```
-----  
ADC Value           : 2974  
Moisture Percentage: 27%  
Pump Status         : ON  
-----
```

Low Moisture content in the soil and high ADC value, Pump is in ON State

Conclusion

The project successfully demonstrated an automated irrigation system using ESP32. Soil moisture was continuously monitored, threshold logic controlled the relay reliably, and the system responded correctly by switching the pump ON or OFF based on moisture conditions.